

GROMACS Test System: Chignolin

This repository contains a simple GROMACS workflow for simulating the Chignolin peptide (PDB ID: 5AWL). The workflow includes system preparation, solvation, ion addition, energy minimization, and production run.

1. Access compute node

First, access a short compute node and load GROMACS:

```
srunk --pty -p short -c 32 --time=03:00:00 bash
module load gromacs
```

2. Generate topology

Generate the processed structure and topology using `pdb2gmx`.

Here we use TIP3P water and AMBER03 force field for protein with AMBER94 nucleic parameters (Duan et al., *J. Comput. Chem.* **24**, 1999–2012, 2003).

```
gmx pdb2gmx -f 5AWL.pdb -o protein_processed.gro -water tip3p
```

When prompted, select **1 "AMBER03 protein, nucleic AMBER94"**.

Input

- 5AWL.pdb : Initial protein structure

Output

- protein_processed.gro : Processed structure
 - topol.top : Topology file
 - posre.itp : Position restraints
-

3. Define unit cell & solvate system

Create simulation box

```
gmx editconf -f protein_processed.gro -o protein_newbox.gro -c -d 1.0 -bt cubic
```

Add solvent

```
gmx solvate -cp protein_newbox.gro -cs spc216.gro -o protein_solv.gro -p topol.top
```

Output

- protein_newbox.gro : Boxed system
 - protein_solv.gro : Solvated system
 - Updated topol.top
-

4. Add ions

Prepare ion input file:

```
gmx grompp -f input/ions.mdp -c protein_solv.gro -p topol.top -o ions.tpr
```

Insert ions:

```
gmx genion -s ions.tpr -o protein_solv_ions.gro -p topol.top -pname NA -nname CL -neutral
```

When prompted, select **group 13 "SOL"**.

Output

- protein_solv_ions.gro : Neutralized system
 - Updated topol.top
 - ions.tpr
-

5. Energy minimization

Before running dynamics, minimize the system to remove bad contacts.

Run minimization

```
gmx grompp -f input/minim.mdp -c protein_solv_ions.gro -p topol.top -o em.tpr  
gmx mdrun -deffnm em -ntmpi 4 -ntomp 4 -v
```

Analyze energy

```
gmx energy -f em.edr -o potential.xvg
```

At the prompt:

10 0

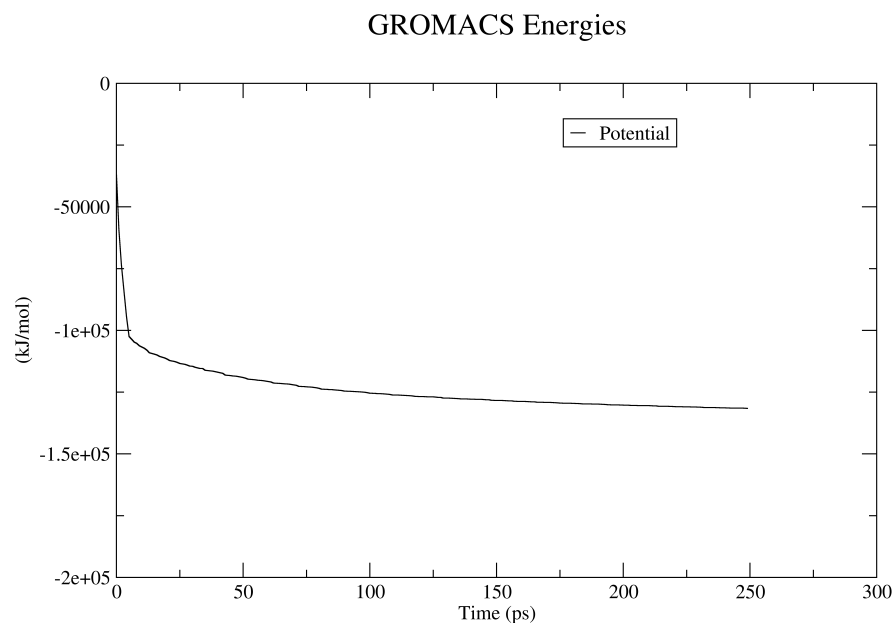
- 10 → Potential energy
- 0 → Exit

Output

- em.gro : Minimized structure
- em.edr : Energy file

- `potential.xvg` : Energy profile

Potential energy plot



6. Equilibration

Energy minimization provides a relaxed starting structure, but the system is not yet ready for production dynamics. The solvent and ions must be equilibrated around the protein.

If unrestrained dynamics are performed at this stage, the system may become unstable because the solvent is not yet properly adapted to the solute. Therefore, equilibration is performed to gradually bring the system to the desired temperature and stabilize interactions.

Equilibration is typically done in two phases:

- **NVT (constant Number of particles, Volume, Temperature)** → temperature stabilization
 - **NPT (constant Number of particles, Pressure, Temperature)** → density/pressure stabilization
-

6.1 NVT equilibration

Prepare input file

```
gmx grompp -f input/nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr
```

Run equilibration

```
gmx mdrun -deffnm nvt -v -ntmpi 4 -ntomp 4
```

6.2 Temperature analysis

Analyze temperature stability using the energy module:

```
gmx energy -f nvt.edr -o temperature.xvg
```

Interactive selection

At the prompt, enter:

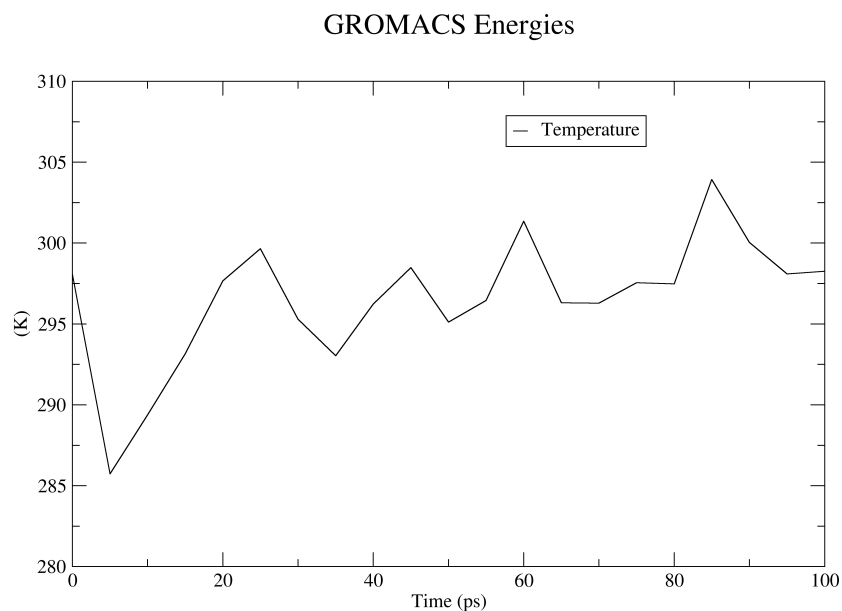
```
16 0
```

- 16 → Temperature of the system
 - 0 → Exit selection
-

Output

- `nvt.tpr` : Input file for simulation
 - `nvt.gro` : Equilibrated structure
 - `nvt.edr` : Energy file
 - `temperature.xvg` : Temperature evolution
-

Temperature plot



7. NPT Equilibration

After completing the NVT step, the system temperature is stabilized. The next step is to stabilize the pressure and density of the system before production dynamics.

This is done using the **NPT ensemble (constant Number of particles, Pressure, and Temperature)**, also known as the isothermal–isobaric ensemble. This ensemble closely resembles experimental conditions.

7.1 Run NPT equilibration

Prepare input file

```
gmx grompp -f input/npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr
```

Run simulation

```
gmx mdrun -deffnm npt
```

7.2 Pressure analysis

Analyze pressure fluctuations using the GROMACS energy module:

```
gmx energy -f npt.edr -o pressure.xvg
```

Interactive selection

At the prompt, enter:

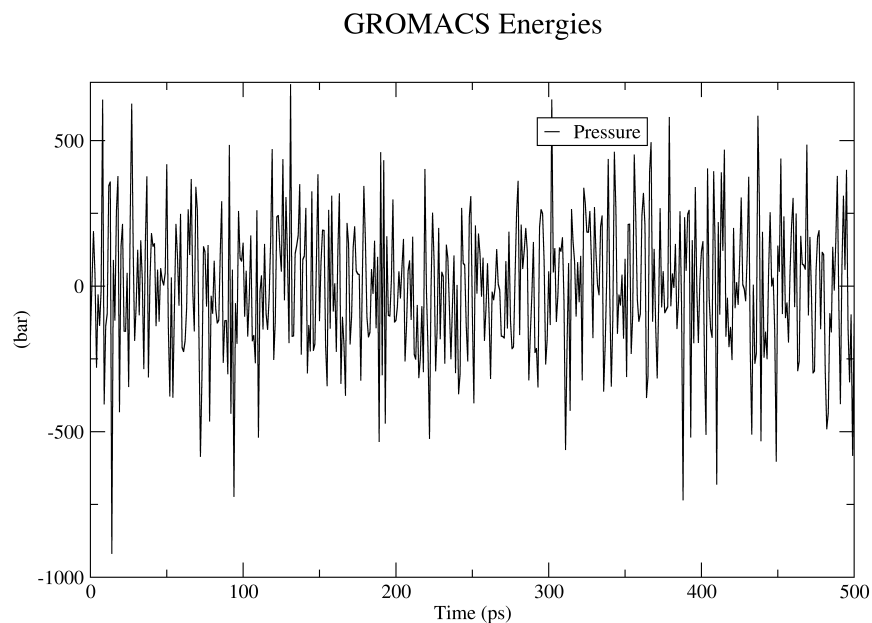
```
16 0
```

- 17 → Pressure of the system
 - 0 → Exit selection
-

Output

- `npt.tpr` : Input file for NPT run
 - `npt.gro` : Equilibrated structure
 - `npt.cpt` : Checkpoint file
 - `npt.edr` : Energy file
 - `pressure.xvg` : Pressure evolution
-

Pressure plot



8. Production Molecular Dynamics (MD)

After completing both equilibration steps (NVT and NPT), the system is fully equilibrated at the desired temperature and pressure. At this stage, position restraints are removed and the system is ready for **production MD**, where data is collected for analysis.

Here, we perform a **1 ns molecular dynamics simulation**.

8.1 Prepare MD input

```
gmx grompp -f input/md.mdp -c npt.gro -t npt.cpt -p topol.top -o md_0_1.tpr
```

8.2 Run production MD

```
gmx mdrun -deffnm md_0_1
```

Output files

- md_0_1.tpr : Input file for production MD
 - md_0_1.xtc : Trajectory file
 - md_0_1.gro : Final structure
 - md_0_1.edr : Energy file
 - md_0_1.log : Log file
 - md_0_1.cpt : Checkpoint file
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Notes

- This step generates the trajectory used for all downstream analysis.
- Ensure sufficient sampling time depending on the system (1 ns is used here for a test run).

Performance Summary

Core time (s): 11223.211
Wall time (s): 701.506
Parallel efficiency (%): 1599.9

Performance

- ns/day: 123.164
- hour/ns: 0.195

Notes

- Core time = total CPU time across all cores
- Wall time = actual elapsed time
- ns/day = simulation speed
- hour/ns = time per nanosecond

Interpretation

- **ns/day = 123.164**
→ The simulation speed is ~123 nanoseconds per day of wall-clock time.
- **hour/ns = 0.195**
→ Each nanosecond of simulation takes ~0.195 hours (~11.7 minutes).

Note

- Performance depends on system size, force field, CPU architecture, and MPI/OpenMP balance
- This benchmark indicates a highly efficient run for a small system like Chignolin.